

**Zubair Mughal**

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

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**AI**

COURSE 05 OF 12  
CURRICULUM GUIDE

# Agentic AI for Teens

Building AI systems that plan, use tools & act

GRADE BAND

**Grade 9–12**

DIFFICULTY

**Intermediate–Advanced**

DURATION

**10 weeks · 2 sessions/week**

## ABOUT THIS COURSE

## Course Overview

Beyond chatbots: this course explores how autonomous AI agents reason, plan, call tools and complete multi-step tasks — and has students build their own, responsibly.

|                                 |                                            |                                               |
|---------------------------------|--------------------------------------------|-----------------------------------------------|
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|---------------------------------|--------------------------------------------|-----------------------------------------------|

Prerequisites: Python for Kids & Teens; Generative AI for Teens recommended.

## LEARNING OUTCOMES

## What Students Will Learn

- Understand what makes an AI system “agentic” versus a simple chatbot
- Understand how agents plan, use tools/APIs, and take actions
- Design prompts and workflows for reliable agent behavior
- Build a simple tool-using AI agent with Python
- Recognize the limitations, risks and responsible-use considerations of AI agents
- Evaluate and iterate on agent behavior using test cases

## WHAT WE'LL USE

## Tools & Technologies

Python   An LLM API   Function / tool calling   Basic automation scripting

## CURRICULUM

## Module-by-Module Breakdown

| MODULE                          | FOCUS                                     |
|---------------------------------|-------------------------------------------|
| 01 · What Are AI Agents?        | Agents vs. chatbots, real-world examples. |
| 02 · How Agents Think           | Planning, reasoning steps and tool use.   |
| 03 · Connecting Agents to Tools | APIs and function-calling basics.         |

| MODULE                            | FOCUS                                           |
|-----------------------------------|-------------------------------------------------|
| 04 · Building a Simple Agent      | First hands-on agent project.                   |
| 05 · Multi-Step Tasks             | Chaining actions and handling errors.           |
| 06 · Responsible AI & Limitations | Bias, hallucination, safe use, human oversight. |
| 07 · Testing & Improving Agents   | Evaluating outputs and iterating on prompts.    |
| 08 · Capstone Agent Project       | A self-designed, task-completing agent.         |

### LEARNING BY BUILDING

## Projects Students Will Build

- AI Study Helper Agent — Answers questions and summarizes notes.
- Task Automation Agent — Completes a defined multi-step task.
- Capstone Agent — A custom AI agent solving a problem the student defines.

### BEYOND THE SYNTAX

## Skills Developed

Prompt engineering   API integration   Systems thinking   Responsible AI use

### PORTFOLIO OUTCOME

A working AI agent demo with documentation explaining its design, behavior and limitations.

### RECOMMENDED NEXT STEP

System Design for Teens